LESSON 1 NOTES - MATRICES

SECTION 1 - MATRICES BASICS

1.0: DEFINITION - A MATRIX

We review earlier learnt ideas about matrices (in High School) and highlight the algebraic operations defined on them.

INTRODUCTION

A *matrix A over a field K* or, simply, a *matrix A* (when K is implicit) is a rectangular array of scalars usually presented in the following form:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

The *rows* of such a matrix A are the m horizontal lists of scalars:

$$(a_{11}, a_{12}, \dots, a_{1n}), \quad (a_{21}, a_{22}, \dots, a_{2n}), \quad \dots, \quad (a_{m1}, a_{m2}, \dots, a_{mn})$$

and the *columns* of A are the n vertical lists of scalars:

$$\begin{bmatrix} a_{11} \\ a_{21} \\ \dots \\ a_{m1} \end{bmatrix}, \quad \begin{bmatrix} a_{12} \\ a_{22} \\ \dots \\ a_{m2} \end{bmatrix}, \quad \dots, \quad \begin{bmatrix} a_{1n} \\ a_{2n} \\ \dots \\ a_{mn} \end{bmatrix}$$

Note that the element a_{ij}, called the *ij-entry* or *ij-element*, appears in row i and column j.

We frequently denote such a matrix by simply writing $A = [a_{ij}]$.

A matrix with m rows and n columns is called an m *by* n matrix, written $m \times n$. The pair of numbers m and n is called the *size* of the matrix. Two matrices A and B are *equal*, written $A = B$, if they have the same size and if corresponding elements are equal. Thus, the equality of two $m \times n$ matrices is equivalent to a system of mn equalities, one for each corresponding pair of elements.

A matrix with only one row is called a *row matrix* or *row vector*, and a matrix with only one column is called a *column matrix* or *column vector*. A matrix whose entries are all zero is called a *zero matrix* and will usually be denoted by 0.

Matrices whose entries are all real numbers are called *real matrices* and are said to be *matrices over* **R**. Analogously, matrices whose entries are all complex numbers are called *complex matrices* and are said to be *matrices over* **C**. This text will be mainly concerned with such real and complex matrices.

EXAMPLES

$A = \begin{bmatrix} 1 & -4 & 5 \\ 0 & 3 & -2 \end{bmatrix}$ is a 2×3 matrix.

Its rows are $(1, -4, 5)$ and $(0, 3, -2)$

its columns are $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ -2 \end{bmatrix}$

(b) The 2×4 zero matrix is the matrix $0 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Definition 1 If A and B are both $n \times m$ matrices then we say that $A = B$ provided corresponding entries from each matrix are equal. Or in other words, $A = B$ provided $a_{ij} = b_{ij}$ for all i and j.

Matrices of different sizes cannot be equal.

Example 1 Consider the following matrices.

$$A=\begin{bmatrix} -9 & 123 \\ 3 & -7 \end{bmatrix} \qquad B=\begin{bmatrix} -9 & b \\ 3 & -7 \end{bmatrix} \qquad C=\begin{bmatrix} -9 \\ 3 \end{bmatrix}$$

For these matrices we have that $A \neq C$ and $B \neq C$ since they are different sizes and so can't be equal. The fact that C is essentially the first column of both A and B is not important to determining equality in this case. The size of the two matrices is the first thing we should look at in determining equality.

Next, $A = B$ provided we have $b = 123$. If $b \neq 123$ then we will have $A \neq B$.

EXAMPLE

Find x, y, z, t such that

$$\begin{bmatrix} x+y & 2z+t \\ x-y & z-t \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 1 & 5 \end{bmatrix}$$

By definition of equality of matrices, the four corresponding entries must be equal. Thus,

$$x+y=3, \qquad x-y=1, \qquad 2z+t=7, \qquad z-t=5$$

Solving the above system of equations yields $x = 2$, $y = 1$, $z = 4$, $t = -1$.

1.11: Denotation of a matrix

Matrices are denoted using upper case letters **A, B, C, D**,….etc.

If the elements are letters themselves, they are inputted in the matrix using lower case letters.

Definition 1.1: Transpose of a Matrix

The transpose of a matrix $\mathbf{A}$, denoted by $\mathbf{A}^T$, is the matrix obtained by writing the collumns of $\mathbf{A}$, in order, as rows.

EXAMPLE

$$\text{For example, } \mathbf{A}^T = \begin{bmatrix} 3 & -7 & 8 \\ 4 & 5 & -2 \end{bmatrix}^T = \begin{bmatrix} 3 & 4 \\ -7 & 5 \\ 8 & -2 \end{bmatrix}$$

EXERCISE

$$\text{Find } \mathbf{C}^T, \mathbf{D}^T, \text{ given } \mathbf{C} = \begin{bmatrix} 3 & -4 & 5 \\ -2 & 7 & 2 \\ 8 & -7 & 6 \end{bmatrix} \text{ and } \mathbf{D} = \begin{bmatrix} 5 & -6 & 0 \\ 1 & 9 & -4 \end{bmatrix}.$$

THEOREM: PROPERTIES OF THE TRANSPOSE

Let **A** and **B** be matrices and let k be a scalar. Then whenever the sum **A** + **B** and product **AB** are defined,

Let **A** and **B** be matrices and let k be a scalar. Then whenever the sum **A** + **B** and product **AB** are defined,

1. $(A^T)^T = A$
2. $(A+B)^T = A^T + B^T$ *and* $(A-B)^T = A^T - B^T$
3. $(kA)^T = kA^T$ Where k is a scalar
4. $(AB)^T = B^T A^T$

1.2: SOME SPECIAL MATRICES

We list a few special types of matrices.

a. The Zero Matrix

A matrix whose entries are all zero is called a zero matrix. Its denoted by **O**.

$$\mathbf{O}_{2x3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ is a zero matrix of order 2 x 3.}$$

b. A Square Matrix

A ***square matrix*** is a matrix with the same number of rows as columns.

An *n x n* square matrix is said to be of order *n*.

Exercise

Write down a square matrix of order (i) 3 (ii) 5 (iii) order *n*.

Definition 1.2: The Diagonal of a square matrix

The diagonal, or the main diagonal of a square matrix **A** consists of all the elements with the same subscripts, that is $a_{11}, a_{22}, a_{33}, \cdots, a_{nn}$.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}$$

c. A Diagonal Matrix

A square matrix in which every element except the principal / main diagonal elements is zero is called a Diagonal Matrix.

Alternatively;

A square matrix $\mathbf{D} = [d_{ij}]_{n\times n}$ will be called a Diagonal Matrix if $d_{ij} = 0$ whenever i is not equal to j.

d. An Identity Matrix

A Diagonal Matrix $\mathbf{A}$ is called an identity matrix if $a_{ij} = 1$ for all $i = j$.
An identity matrix of order n is normally denoted by I_n.

Exercise
Write down an Identity matrix of order (i) 2 (ii) 5 (iii) order *n*.

e. Triangular Matrices

(i) An Upper Triangular Matrix

A square matrix $\mathbf{A}$ is called an ***upper triangular*** if all the entries below the main diagonal are equal to zero.
Generic upper triangular matrices of order 2, 3 and 4 are below:

$$\text{order } 2 \begin{bmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{bmatrix} \qquad \text{order } 3 \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

$$\text{order } 4 \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

(ii) A Lower Triangular Matrix

A square matrix **A** is called an ***upper triangular*** if all the entries below the main diagonal are equal to zero. Generic upper triangular matrices of order 2, 3 and 4 are below:

$$\text{order } 2 \begin{bmatrix} 0 & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \qquad \text{order } 3 \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{order } 4 \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

Exercise

Classify as upper triangular or lower triangular matrix.

$$\mathrm{D} = \begin{bmatrix} 7 & -3 \\ 0 & 0 \end{bmatrix} \qquad \mathrm{A} = \begin{bmatrix} 0 & -9 \\ 2 & 5 \end{bmatrix}$$

$$\mathrm{B} = \begin{bmatrix} 7 & -3 \\ 0 & 6 \end{bmatrix} \qquad \mathrm{C} = \begin{bmatrix} 0 & 0 \\ 2 & 5 \end{bmatrix}$$

$$\mathbf{F} = \begin{bmatrix} 6 & 0 & -4 \\ 0 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix} \qquad \mathbf{E} = \begin{bmatrix} 1 & 5 & -4 \\ 0 & 8 & -2 \\ 0 & 0 & 7 \end{bmatrix}$$

$$\mathbf{M} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & -2 & 0 \end{bmatrix} \qquad \mathbf{H} = \begin{bmatrix} 9 & 0 & 0 \\ 2 & -4 & 0 \\ 7 & -2 & 8 \end{bmatrix}$$

f. Symetric Matrices

i) A Symetric Matrix.

A square matrix $\mathbf{A}$ is called ***Symetric*** if $\mathbf{A} = \mathbf{A}^T$, that is, if $\mathbf{A}$ equals its Transpose $\mathbf{A}^T$

ii) A Skew – Symetric Matrix.

A square matrix $\mathbf{A}$ is called ***Skew - Symetric*** if $\mathbf{A} = -\mathbf{A}^T$.

Exercise

Classify as a Symetric or Skew - Symetric Matrix.

$$\mathbf{B} = \begin{bmatrix} 5 & 0 \\ 0 & 6 \end{bmatrix} \qquad \mathbf{M} = \begin{bmatrix} 2 & -7 \\ -7 & 11 \end{bmatrix} \qquad \mathbf{F} = \begin{bmatrix} 2 & -7 & 6 \\ -7 & 3 & 5 \\ 6 & 5 & -4 \end{bmatrix}$$

$$\mathbf{H} = \begin{bmatrix} 9 & 0 & 0 \\ 2 & -4 & 0 \\ 7 & -2 & 8 \end{bmatrix} \qquad \mathbf{D} = \begin{bmatrix} 8 & 1 \\ -1 & 4 \end{bmatrix} \qquad \mathbf{G} = \begin{bmatrix} 0 & 3 & -4 \\ -3 & 0 & -2 \\ 4 & 2 & 0 \end{bmatrix}$$

$$\mathbf{M} = \begin{bmatrix} 2 & -7 \\ -7 & 11 \end{bmatrix} \qquad \mathbf{F} = \begin{bmatrix} 2 & -7 & 6 \\ -7 & 3 & 5 \\ 6 & 5 & -4 \end{bmatrix} \qquad \mathbf{C} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 7 \\ -7 & 0 \end{bmatrix} \qquad \mathbf{E} = \begin{bmatrix} 1 & 5 & -4 \\ 0 & 8 & -2 \\ 0 & 0 & 7 \end{bmatrix}$$

g. Normal Matrix

A real matrix A is *normal* if it *commutes* with its transpose A^T—that is, if $AA^T = A^TA$. If A is symmetric, orthogonal, or skew-symmetric, then A is normal. There are also other normal matrices.

Example

Let $A = \begin{bmatrix} 6 & -3 \\ 3 & 6 \end{bmatrix}$

Show that matrix A is normal.

$$AA^T = \begin{bmatrix} 6 & -3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 6 & 3 \\ -3 & 6 \end{bmatrix} = \begin{bmatrix} 45 & 0 \\ 0 & 45 \end{bmatrix}$$ **and**

$$A^TA = \begin{bmatrix} 6 & 3 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} 6 & -3 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 45 & 0 \\ 0 & 45 \end{bmatrix}$$

h. Orthogonal Matrix

A real matrix $\mathbf{A}$ is ***Orthogonal*** if $\mathbf{A}^T = \mathbf{A}^{-1}$. In other words, $\mathbf{A}$ is ***Orthogonal*** if $\mathbf{A}^T = \mathbf{A}^{-1}$. Thus, $\mathbf{A}$ must necessarily be square and invertible.

Examples

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \qquad \mathbf{C} = \begin{bmatrix} \frac{1}{9} & \frac{8}{9} & -\frac{1}{9} \\ \frac{4}{9} & -\frac{4}{9} & -\frac{1}{9} \\ \frac{8}{9} & \frac{1}{9} & \frac{4}{9} \end{bmatrix}$$

Definition 1.2: The Trace of a square matrix

The trace of a matrix **A**, written $tr(A)$, is the sum of the diagonal elements. i.e.if

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2}, the & a_{n3} & \cdots & a_{nn} \end{bmatrix}$$

is a square matrix of order n, then

$$tr(A) = a_{11} + a_{22} + a_{33} + \cdots + a_{nn}$$

ALTERNATIVELY,

Definition 4 *If A is an $n \times n$ matrix, then the trace of A denoted by* $\mathrm{tr}(\mathrm{A})$ *is defined as the sum of all the main diagonal elements of A.*

That is, $\mathrm{tr}(\mathrm{A}) = \sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{a_{ii}}$.

Some useful facts about trace operators are given below.

Theorem 2

Let A and B be matrices of appropriate sizes.

1. $\mathrm{tr}(\mathrm{kA}) = \mathrm{k}\,\mathrm{tr}(\mathrm{A})$ *for any* $k \in \mathbf{R}$.

2. $\mathrm{tr}(\mathrm{A} + \mathrm{B}) = \mathrm{tr}(\mathrm{A}) + \mathrm{tr}(\mathrm{B})$.

3. $\mathrm{tr}(\mathrm{AB}) = \mathrm{tr}(\mathrm{BA})$.

4. $\mathrm{tr}(\mathrm{A}^\top) = \mathrm{tr}(\mathrm{A})$.

5. $\mathrm{tr}(\mathrm{A}^\top \mathrm{A}) \geq 0$.

LESSON 1 NOTES - MATRICES

SECTION 2

1.3: MATRIX ALGEBRA / OPERATIONS

1.31: MATRIX ADDITION OR SUBTRACTION

Any of the two definitions below will suffice:

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices with the same size, say $m \times n$ matrices. The *sum* of A and B, written $A + B$, is the matrix obtained by adding corresponding elements from A and B. That is,

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix}$$

Subtraction is similarly done by subtracting corresponding elements of matrices ***A*** and ***B***.

Alternatively we define sum and difference as:

If $\mathbf{A}$ and $\mathbf{B}$ are both $m \times n$ matrices then $\mathbf{A} \pm \mathbf{B}$ is a new $m \times n$ matrix that is found by adding/subtracting corresponding entries from each matrix. Or in other words, $\mathbf{A}_{ij} \pm \mathbf{B}_{ij} = [a_{ij} \pm b_{ij}]$

Note: Matrices of different sizes cannot be added or subtracted.

1.32: SCALAR MULTIPLICATION OF MATRICES

The *product* of the matrix A by a scalar k, written $k \cdot A$ or simply kA, is the matrix obtained by multiplying each element of A by k. That is,

$$kA = \begin{bmatrix} ka_{11} & ka_{12} & \dots & ka_{1n} \\ ka_{21} & ka_{22} & \dots & ka_{2n} \\ \dots & \dots & \dots & \dots \\ ka_{m1} & ka_{m2} & \dots & ka_{mn} \end{bmatrix}$$

If $\mathbf{A}$ is any matrix and k is any number then the product (or scalar multiple), $k\mathbf{A}$, is a new matrix of the same size as $\mathbf{A}$ and it's entries are found by multiplying the original entries of $\mathbf{A}$ by k. In other words $k\mathbf{A}_{ij} = [ka_{ij}]$ for all i and j.

Observe that $A + B$ and kA are also $m \times n$ matrices.

$$-A = (-1)A \qquad \text{and} \qquad A - B = A + (-B)$$

The matrix $-A$ is called the *negative* of the matrix A, and the matrix $A - B$ is called the *difference* of A and B. The sum of matrices with different sizes is not defined.

Basic properties of matrices under the operations of matrix addition and scalar multiplication are as in the theorem below:

THEOREM 2.1:

Consider any matrices A, B, C (with the same size) and any scalars k and k'. Then

(i) $(A + B) + C = A + (B + C)$,

(ii) $A + 0 = 0 + A = A$,

(iii) $A + (-A) = (-A) + A = 0$,

(iv) $A + B = B + A$,

(v) $k(A + B) = kA + kB$,

(vi) $(k + k')A = kA + k'A$,

(vii) $(kk')A = k(k'A)$,

(viii) $1 \cdot A = A$.

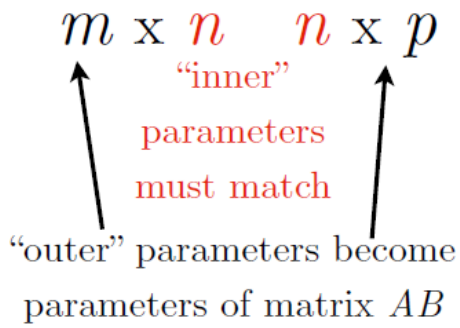
Note first that the 0 in (ii) and (iii) refers to the zero matrix.

1.33: MULTIPLICATION OF MATRICES *A* X *B* / PRODUCT OF 2 MATRICES *A* AND *B*

What Matrices Can Be Multiplied Together / Compatible Matrices

What sizes of matrices can be multiplied together?

For m x n matrix A and n x p matrix B, the matrix product AB is an m x p matrix.



This definition tells us that the product AB is only defined if A (*i.e.* the first matrix listed in the product) has the same number of columns as B (*i.e.* the second matrix listed in the product) has rows.
If the number of columns of the first matrix listed is not the same as the number of rows of the second matrix listed then the product is not defined.

Checking if two matrices are said to be compatible

An easy way to check that a product is defined is to write down the two matrices in the order that we want to multiply them and underneath them write down the sizes as shown below.

$$\begin{array}{ccccc} A & & B & = & AB \\ n \times p & & p \times m & & n \times m \end{array}$$

If the two inner numbers are equal then the product is defined and the size of the product will be given by the outside numbers.

MULTIPLYING TWO MATRICES *A AND B* / PRODUCT OF 2 MATRICES *A* AND *B*

Definition

Here is the formal definition of matrix multiplication:

Definition:

Let $\mathbf{A} = [a_{ij}]_{m\times p}$ be an $m \times p$ matrix and $\mathbf{B} = [b_{ij}]_{p\times r}$ be an $p \times r$ matrix. Then we define the product $\mathbf{C} = \mathbf{A} \times \mathbf{B} = [a_{ij}] \times [b_{ij}] = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots a_{ip}b_{pj}$

Notice:

i) $a_{i1};\ a_{i2};\ a_{i3};\ \cdots\ ;\ a_{ip}$ are the elements of row i of $\mathbf{A}$, while $b_{1j};\ b_{2j};\ b_{3j};\ \cdots\ ;\ b_{pj}$ are the elements of column j of $\mathbf{B}$.

ii. The element in the first row of **A x B** is the inner product of the first row of **A** with the column matrix **B**.

iii. The element in the second row of **A x B** is the inner product of the second row of **A** with the column matrix **B**; and so on….

iv. For matrix multiplication to be defined, the number of columns of **A** must be the same as the number of rows of **B**.

EXAMPLE

$$\text{Let } \mathbf{A} = \begin{bmatrix} -3 & 2 \\ 5 & -8 \\ -7 & 4 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 6 & 1 \\ -9 & 2 \end{bmatrix}$$

$$\text{Then } \mathbf{A} \times \mathbf{B} = \begin{bmatrix} -3 & 2 \\ 5 & -8 \\ -7 & 4 \end{bmatrix} \times \begin{bmatrix} 6 & 1 \\ -9 & 2 \end{bmatrix} = \begin{bmatrix} (-3)(6)+(2)(-9) & (-3)(1)+(2)(2) \\ (5)(6)+(-8)(-9) & (5)(1)+(-8)(2) \\ (-7)(6)+(4)(-9) & (-7)(1)+(4)(2) \end{bmatrix}$$

$$= \begin{bmatrix} -18+-18 & -3+4 \\ 30+72 & 5+-16 \\ -42+-36 & -7+8 \end{bmatrix} = \begin{bmatrix} -36 & 1 \\ 102 & -11 \\ -78 & 1 \end{bmatrix}$$

Basic properties of matrices multiplication are contained in the theorem below.

THEOREM 2.2:

Let A, B, C be matrices. Then, whenever the products and sums are defined,

(i) $(AB)C = A(BC)$ (associative law),

(ii) $A(B+C) = AB + AC$ (left distributive law),

(iii) $(B+C)A = BA + CA$ (right distributive law),

(iv) $k(AB) = (kA)B = A(kB)$, where k is a scalar.

Note: Matrix multiplication is not commutative, i.e:

In general, $AB \neq BA$.

MATRICES AND (SIMPLE) LINEAR SYSTEMS (ORDERS 2X2 AND 3X3)

Matrix multiplication also gives us a very nice and compact way of writing systems of linear equations. We will start with a simple liear system of order 2x2 and 3x3.

Note: these will later generalized to a linear system of any order m x n.

Consider a linear system of order 2 (two linear equations in two unknowns). i.e.

$$ax + by = e$$
$$cx + dy = f$$

Now instead of thinking of these as a set of equations, let's think of each side as a vector of size $n \times 1$ as follows:

$$\mathbf{B} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix} = \begin{bmatrix} e \\ f \end{bmatrix}.$$

With a little working, we recognize that the left hand side can be written as the following product:

$$\mathbf{B} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} e \\ f \end{bmatrix}.$$

We will use this fact to develop some methods of solving linear systems (using determinants and inverses, for example)

1.4: WORD PROBLEMS / USE OF MATRICES IN OUR DAILY LIVING

SET A QNS

1. There are 5,500 men, women and children altogether at the swimming pool. There are twice as many women as men and four times as many children as women. How many men, women and children are at the swimming pool ?
2. If one side of the triangle increases by 11 cm and the other side decreases by the same value, we get an equilateral triangle. When the first side is multiplied by four, it is 10 cm longer than three times the third side. Find the lengths of the triangle's sides.

3. If one dimension of the cuboid increases by 1 cm, the surface area of the cuboid increases by 54 cm^2. If the second dimension of the cuboid increases by 2 cm, the surface area of the cuboid increases by 96 cm^2. If the third dimension of the cuboid increases by 3 cm, its surface area increases by 126 cm^2. Find the dimensions of the cuboid.
4. The four-digit number has a digit sum of 20. The sum of its last two digits is equal to its second digit increased by 5, the sum of its outer digits equals to its second digit decreased by 3. If we write the digits of the number in reverse order, the number increases by 2,178. Find the number.
5. A cylinder made out of the yellow copper has the weight of 6.297 kg and the volume of 750 cm^3. How many kilograms of copper and how many kilograms of zinc the cylinder contains ? (The density of copper is 8,900 kg/m^3, the density of zinc is 7,100 kg/m^3.)
6. By adding 30 kg of pure tin we have to prepare a bronze alloy made out of 75% of copper and 25% of tin. For our disposal we have a bronze alloy B made out of 85% of copper and 15% of tin. How many kilograms of the alloy B do we have to use ? How many kilograms of the alloy A will we obtain ?
7. Two containers contain a water of different temperatures. If we mix 240 g of water from the first container with 260 g of water from the second container, the resulting water temperature will be 52°C. If we mix 180 g of water from the first container with 120 g of water from the second container, the resulting water temperature will be 46°C. What is the temperature of water in the containers ?
8. How many grams of an 80% solution and how many grams of a 54% solution do we have to mix in order to obtain 100 g of a 60% solution ? (% is meant as by weight)
9. How many liters of a 30% solution and how many liters of a 50% solution do we have to mix in order to obtain 25 liters of a 42% solution? (% is meant as by volume)
10. The sulfuric acid consists of hydrogen, oxygen and sulfur, wherein the weight ratio of the hydrogen and the sulfur is 1:16 and the weight ratio of the oxygen and the sulfur is 2:1. How many of each element contains 1,323 grams of the acid ?
11. How many kilograms of iron and how many kilograms of sulfur contains 100 kg of FeS, if the relative atomic weight of iron is 56 and the relative atomic weight of sulfur is 32 ?

SET A ANSWERS

1. There are 500 men, 1,000 women and 4,000 children at the swimming pool.

2. The triangle's sides are 43 cm, 65 cm and 54 cm long.

3. Dimensions of the cuboid are 9 cm, 12 cm and 15 cm.

4. The wanted number is 1,793.

5. The cylinder contains 4.806 kg of copper and 1.491 kg of zinc.

6. We have to use 225 kg of the alloy B, and by that we will obtain 255 kg of the alloy A.

7. The temperature of water in the containers is 26°C and 76°C.

8. We have to mix 23.077 grams of an 80% solution and 76.923 grams of a 54% solution.

9. We have to mix 10 liters of a 30% solution and 15 liters of a 50% solution.

10. 1,323 grams of the sulfuric acid contains 27 grams of hydrogen, 432 grams of sulfur and 864 grams of oxygen.

11. 100 kg of FeS contains 63.64 kg of iron and 36.36 kg of sulfur.

BOX-OFFICE RECEIPTS A theater has a seating capacity of 900 and charges $4 for children, $6 for students, and $8 for adults. At a certain screening with full attendance, there were half as many adults as children and students combined. The receipts totaled $5600. How many children attended the show?

MIXTURES The Coffee Shoppe sells a coffee blend made from two coffees, one costing $5/lb and the other costing $6/lb. If the blended coffee sells for $5.60/lb, find how much of each coffee is used to obtain the desired blend. Assume that the weight of the blended coffee is 100 lb.

MANAGEMENT DECISIONS The management of Hartman Rent-A-Car has allocated $1.5 million to buy a fleet of new automobiles consisting of compact, intermediate-size, and full-size cars. Compacts cost $12,000 each, intermediate-size cars cost $18,000 each, and full-size cars cost $24,000 each. If Hartman purchases twice as many compacts as intermediate-size cars and the total number of cars to be purchased is 100, determine how many cars of each type will be purchased. (Assume that the entire budget will be used.)

INVESTMENTS Kelly Fisher has a total of $30,000 invested in two municipal bonds that have yields of 8% and 10% interest per year, respectively. If the interest Kelly receives from the bonds in a year is $2640, how much does she have invested in each bond?

MIXTURE PROBLEM—FERTILIZER Lawnco produces three grades of commercial fertilizers. A 100-lb bag of grade-A fertilizer contains 18 lb of nitrogen, 4 lb of phosphate, and 5 lb of potassium. A 100-lb bag of grade-B fertilizer contains 20 lb of nitrogen and 4 lb each of phosphate and potassium. A 100-lb bag of grade-C fertilizer contains 24 lb of nitrogen, 3 lb of phosphate, and 6 lb of potassium. How many 100-lb bags of each of the three grades of fertilizers should Lawnco produce if 26,400 lb of nitrogen, 4900 lb of phosphate, and 6200 lb of potassium are available and all the nutrients are used?

DIET PLANNING A dietitian wishes to plan a meal around three foods. The percentage of the daily requirements of proteins, carbohydrates, and iron contained in each ounce of the three foods is summarized in the following table:

	Food I	Food II	Food III
Proteins (%)	10	6	8
Carbohydrates (%)	10	12	6
Iron (%)	5	4	12

Determine how many ounces of each food the dietitian should include in the meal to meet exactly the daily requirement of proteins, carbohydrates, and iron (100% of each).